

1 Voting Theory

1.2 Pairwise Comparison Method

Example 1.2.1 Let’s revisit an example from last time.

	1	3	3	3
1st Choice	A	A	O	H
2nd Choice	O	H	H	A
3rd Choice	H	O	A	O

Review plurality method. and go over all pairwise comparisons. Is there a Condorcet Candidate? Did they win the election? Review definitions of Condorcet Candidate and the Condorcet Criteria if needed.

Definition 1.2.2 (Pairwise Comparison Method) Given a preference schedule for an election, perform all one-on-one comparisons (eliminate all but 2 candidates at a time). A candidate recieves a point for each comparison the candidate wins. If there is a tie, each candidate gets half a point. At the end, the candidate with the most points wins.

Example 1.2.3 Suppose we must award one of four students, labeled A , B , C , and D , a scholarship. The comitee submits the following preference schedule. With the pairwise comparison method, which student will recieve the scholarship? election?

	5	5	6	4
1st	D	A	C	B
2nd	A	C	B	D
3rd	C	B	D	A
4th	B	D	A	C

Example 1.2.4 Your turn! Using the pairwise comparison method, which candidate wins the following election?

	3	4	4	6	2	1
1st	B	C	B	D	B	E
2nd	C	A	D	C	E	A
3rd	A	D	C	A	A	D
4th	D	B	A	E	C	B
5th	E	E	E	B	D	C

(Hint: there are 10 comparisons to make on this one)

Question: Does the pairwise comparison method satisfy the Condorcet Criterion?

Example 1.2.5 Using the pairwise comparison method, which candidate wins the following election?

	13	14	11	9
1st	B	D	B	C
2nd	C	C	A	A
3rd	A	A	D	D
4th	D	B	C	B

So what is wrong with this method?

Example 1.2.6 Suppose in one of the previous examples, (recall that C was the winner), the candidate D is discovered to be ineligible for the scholarship due to failing a class last semester. With D gone, the comittee decides to recount the votes using the same method. Is the outcome the same?

	5	5	6	4
1st	D	A	C	B
2nd	A	C	B	D
3rd	C	B	D	A
4th	B	D	A	C

Definition 1.2.7 (Independence of Irrelevant Alternatives (IIA) Criterion) If a non-winning candidate is removed from the ballot, it should not change the winner of the election.

Note that the pairwise comparison method can violate this criterion.